

Sanjoli Narang

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LinkedIn

Google Scholar

Education

- Ongoing **Massachusetts Institute of Technology**, *Ph.D., Electrical Engineering and Computer Science*
GPA: **5.0/5.0** Advisor: Prof. Manya Ghobadi
- 2023–24 **Massachusetts Institute of Technology**, *M.S., Electrical Engineering and Computer Science*
GPA: **5.0/5.0** Advisor: Prof. Manya Ghobadi
- 2017–21 **IIT Bombay**, *B.Tech., Electrical Engineering with Honors; minor in Computer Science*
GPA: **9.78/10** Department rank: **6/140**
Ramesh Chandra Sinha Academic Excellence Award: **highest-ranked female undergraduate, Class of 2021.**

Technical Skills

- Languages Python, C/C++, MATLAB; CUDA and CUTLASS (basic).
- Systems Linux kernel development, TCP stack, eBPF, P4/Tofino, ns-3.
- Libraries JAX, PyTorch, vLLM, MaxText, Megatron-LM, DeepSpeed, ColossalAI, TensorFlow, SciPy.
- Hardware RTL design; Verilog, SystemVerilog, VHDL, RDL, UPF.

Selected Research Projects

- 2026 **TpuPilot: TPU Performance Prediction and Optimization**
Systalyze; summer research internship, under submission
Built a CPU-based TPU predictor from compiler schedules and analytical cost models, without hardware profiling. Guided optimizations improved measured training throughput by **1.3–24.6× across six language models.**
- 2026 **MLCC: Congestion Control for Machine-learning Workloads**
MIT; Prof. Manya Ghobadi; accepted at NSDI 2027
Developed Linux kernel congestion control with random-restart annealing for DNN compute/communication overlap. On a 12-GPU testbed, ML-DCQCN achieved up to **1.9× mean and 2.7× 99th-percentile iteration speedups** over DCQCN.
- 2026 **DMart: Market-based Resource Allocation in Datacenters**
MIT; Prof. Venkat Arun, Prof. Manya Ghobadi, under submission
Built multi-objective network scheduling with P4/Tofino auctions and Linux TCP/eBPF bidding agents using dynamic programming. Packet-level simulations showed up to **6× lower deadline miss rates than NDP.**
- 2026 **CXP: Rack-Scale Cache Coherence Beyond RDMA**
MIT; Prof. Manya Ghobadi; under submission
Designed CXL proxies for cross-host cache coherence; simulations showed up to **6× lower average latency than Pilaf** on skewed, read-heavy workloads.
- 2021 **PageRank via Markov Chain Monte Carlo**
IIT Bombay; Prof. Vivek S. Borkar
Analyzed Markov chain Monte Carlo methods for PageRank, including convergence properties and proofs.
- 2020 **Statistical Learning with Low-precision Data**
IIT Bombay; R&D project with Prof. Animesh Kumar
Implemented PCA on single-bit data, preserving classification accuracy with **4× lower** memory use.

Work Experience

- 2026 **Systalyze, Research Intern**
Analytical performance modeling and optimization of ML workloads on TPUs (TpuPilot).
- 2021–23 **Google India, Silicon Digital Design Engineer**
Designed system-on-chip microarchitecture for interrupt handling and debug tracing.
- 2020 **Tyndall National Institute, Ireland, Research Assistant**
Analysis of optical sensing using guided acoustic-wave Brillouin scattering.
- 2020 **Metal Power Analytical, Hardware Engineer**
Developed an FPGA-card hardware model for spectrometer data acquisition.
- 2020 **Larsen & Toubro Defence, IIT Bombay, Overall Design Lead**
Designed remotely operated underwater vehicles for ocean exploration.

Publications

- 2026 **Bring Your Own Objective: Inter-operability of Network Objectives in Datacenters using Markets**
Sanjoli Narang, Anup Agarwal, Venkat Arun, and Manya Ghobadi
Under submission.
- 2026 **TpuPilot: A Principled Approach for Fast & Generic AI Performance Optimization on TPUs**
Sanjoli Narang, Jeremy Carin, Andrew Li, Armin Ghobadi, Sudarsanan Rajasekaran, and Manya Ghobadi
Under submission.
- 2026 **CXP: Rethinking Rack-Scale Coherence Beyond RDMA**
Sanjoli Narang, Weiyang Wang, Ankit Bhardwaj, and Manya Ghobadi
Under submission.
- 2026 **MLCC: A Congestion Control Technique to Accelerate ML Training**
Sanjoli Narang*, Anton Zabreyko*, Sudarsanan Rajasekaran, and Manya Ghobadi
Accepted at NSDI 2027.
- 2025 **Development of a Compton Imager Setup**
Anuraag Arya et al.
Journal of Astrophysics and Astronomy.
- 2024 **MLTCP: A Distributed Technique to Approximate Centralized Flow Scheduling For Machine Learning**
Sudarsanan Rajasekaran, Sanjoli Narang*, Anton Zabreyko*, and Manya Ghobadi
HotNets 2024.
- 2024 **Science with the daksha high energy transients mission**
Varun Bhalerao et al.
Experimental Astronomy 57, 23.
- 2024 **Daksha: on alert for high energy transients**
Varun Bhalerao et al.
Experimental Astronomy 57, 24.
- 2022 **Field-programmable gate array (FPGA) based programmable digital emulator of vibratory micro-electromechanical systems (MEMS) gyroscopes**
Sanjoli Narang and Siddharth Tallur
Review of Scientific Instruments 93(3):035003.
- 2021 **A reconfigurable hardware emulator of MEMS gyroscopes with built-in error source models**
Sanjoli Narang and Siddharth Tallur
IEEE EFTF/IFCS 2021.
- 2021 **Modelling of underwater acoustic networks for source localization in arbitrary bounded reservoirs**
Sanjoli Narang, Aditya Harakare, Nayan Barhate, Andrews Varghese, Aayush Shrivastava, and Ayushi Gupta
IEEE EFTF/IFCS 2021.
*Equal contribution.

Selected Awards and Honors

MIT Jacobs Presidential Fellowship; Ramesh Chandra Sinha Academic Excellence Award, IIT Bombay; Desai Sethi Scholarship; Institute Academic Prize, IIT Bombay.

Undergraduate Research Excellence Awards (URA-01 and URA-02), IIT Bombay; IEEE EFTF-IFCS 2021 Best Student Paper Finalist.

Young Researcher's Prize (AUV-IITB), IEEE Ocean Engineering Society, UT 2021.

JEE Advanced All India Rank 260; JEE Main All India Rank 177.